

measurement branches can arise through voltage-dependent full-charge anchoring rather than continuous SOC propagation. Temperature noise is neutral for DM and small for the SOH-aware pipelines because temperature enters through learned or parameter-dependent paths rather than through the direct charge update.

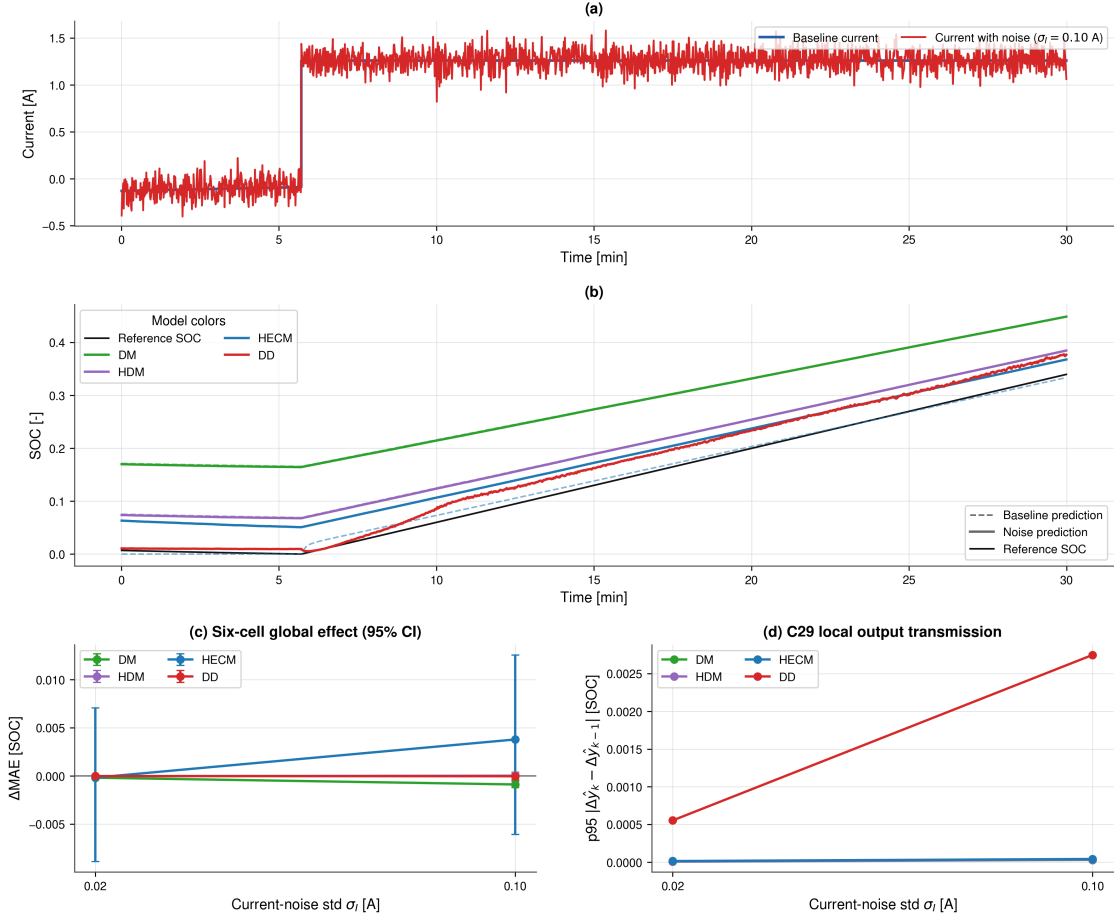


Figure 6.7: Noise sensitivity under additive current noise. Panels (a) and (b) compare current and SOC over a reset-free 30-min example. Panel (c) reports the cell-macro ΔMAE with hierarchical 95% confidence intervals. Panel (d) quantifies local output transmission. DD shows the largest short-term transmission because raw current and its derivative enter the recurrent input directly, while integration in DM and HDM smooths zero-mean sample-level fluctuations.